

Aaditya Kalwani

Aerospace Engineering student focused on machine learning applied to physical engineering systems. Currently researching battery state-of-charge estimation toward publication. Comfortable across the stack: neural networks from scratch in NumPy, C# object-oriented simulations and computational tools for constrained engineering problems. Hands-on UAV 'design and make' experience with Team Bath Drones.

Experience & Technical Projects

Undergraduate Researcher -- Institute for Advanced Automotive Propulsion Systems | Jun '26 - Present

- **Built a machine learning model** to estimate Li-ion cell State of Charge by fusing ultrasound acoustic sensing with transient electric measurements. **First author on paper publication**
- **Implemented a multilayer perceptron from scratch** in NumPy (forward pass, hand-derived backpropagation, training loop) before adopting frameworks to ground the model in first principles
- Cut SOC estimation error to 0.54%, a **10x improvement over the industry method**, by using displacement and ultrasound to correct hysteresis, vs ~6% for coulomb counting alone
- Contributing to the **Energetic (Horizon Europe) project**; paper targeted for peer review in Sep '26

Co-founder & CTO -- Winkit, local food-delivery platform | Aug '26 - Present

- Designed+**built the whole stack**: a customer web app with live Stripe payments linking to **per-restaurant adapters**, placing orders directly onto kitchens' own systems, plus a separate driver app handling shifts, GPS dispatch and real-time order state.
- Removed the need for dedicated point-of-sale hardware, a cost + setup advantage over JustEat
- Launched with real partners, **generating real revenue from 3 restaurants**; onboarding a further 5.

Aerodynamics Division Lead -- Team Bath Drones, Design & Make Competition | Nov '25 – May '26

- **Engineered an autonomous UAV** under strict objectives and material constraints, and **successfully delivered the Critical Design Review** with wing design choices and 400g weight reduction.
- Optimised aerodynamic efficiency: evaluated Clark-Y and NACA0015 profiles for a 30% increase in lift-to-drag ratio; co-led the 'tadpole' fuselage configuration and CAD model

Tech Strategy & AI Engineering Intern -- Fifth-9 (Canary Wharf) | Jun '25 - Present

- **AI Red-Teaming & UAT**: Stress-tested an agentic-AI orchestration platform for enterprise; integrated 6 MCP servers (50+ tools) and ran user-acceptance testing to surface failure modes before release.
- Cortex (cross-LLM memory layer): shipped fixes and built demo tooling, delivering business- and engineer-facing product demos that advanced a live enterprise pilot.
- **Full Stack Development**: Architected and deployed 'PlayDay', a React/Node.js travel-planning engine, integrating the Google Gemini API for real-time, generative itineraries within a 5-day sprint

Founder -- Locus Tutors | Sep '23 - Present

- After managing a tutoring portfolio for 10+ private students, I founded up Locus Tutors: an agency employing Cambridge and Bath Mathematics Students, serving 15 hours of lessons per week.

Selected Engineering Project

- Ultrasonic measurement system (Arduino + HC-SR04, MATLAB): calibrated a low-cost sensor to 1.0mm RMSE (vs 4.3mm spec) via filtering and regression (60% less noise); mapped its $\pm 18^\circ$ beam FOV and drove a state monitor with a variance-based motion-rejection filter for my '**Pringelometer**'.

Education

BEng Aerospace Engineering (Pred. First-Class Honours), University of Bath | Sep '25 - Present

- **Year 1 average: 74%**: Thermofluids (88%), Mechatronics (71%), Design (71%), Mathematics (70%)

Peter Symonds College | A-Levels: A*AAB in Maths, Physics, F. Maths, Comp Sci | Sep '23 - Jun '25

- 2x MEI National Maths Comp. | Silver, Senior Maths Challenge | Bronze, British Physics Olympiad

Selected Technical Skills

- **Languages**: English, Hindi | Python (NumPy, Pandas, ML (PyTorch, LSTM/MLPs)), C# (.NET/OOP), React
- **Engineering Tools**: CAD (Autodesk Inventor), MATLAB, Git+GitHub, Arduino / embedded